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Host a Website on Amazon S3

Shehan Tharushe

Introducing Today's Project!

Project overview

In this project, I will demonstrate how Access Control Lists work in Amazon S3 to control who can access buckets and files, how to enable static website hosting on an S3 bucket and use the bucket website endpoint URL to make it accessible through a browser, and how S3 which stands for Simple Storage Service can be used to store and host files in the cloud. I am doing this project to learn the core features of Amazon S3 including storage, access control, and static website hosting as part of my cloud engineering roadmap with NextWork.ai.

Tools and concepts

Services I used were Amazon S3 to create a bucket, store my website files, and host a static website that is publicly accessible on the internet. Key concepts I learnt include how S3 buckets work as cloud storage spaces and why their names must be globally unique, how to upload HTML and image files into a bucket, how to enable static website hosting to generate a public bucket website endpoint URL, how Access Control Lists control who can access individual files and buckets, how bucket policies use JSON rules to grant or restrict access to an entire bucket, and how making files publicly accessible through permissions is what allows a website to be seen live on the internet

Time, challenges, and wins

This project took me approximately one hour to complete from setting up the S3 bucket to seeing my website live on the internet. The most challenging part was understanding the difference between Access Control Lists and bucket policies, and knowing when to use each one to make my website files publicly accessible without compromising security. It was most rewarding to open my bucket website endpoint URL in the browser and see my website loading live on the internet for the first time, knowing that I had built and hosted it entirely on AWS using Amazon S3

How I Set Up an S3 Bucket

What I did in this step

n this step, we are opening Amazon S3 and creating a bucket, which is a storage space that will hold all the files that make up our website. Think of a bucket like a folder in the cloud where we can store and organize our HTML, CSS, and image files that will later be served as a static website.

How long it took to create the bucket

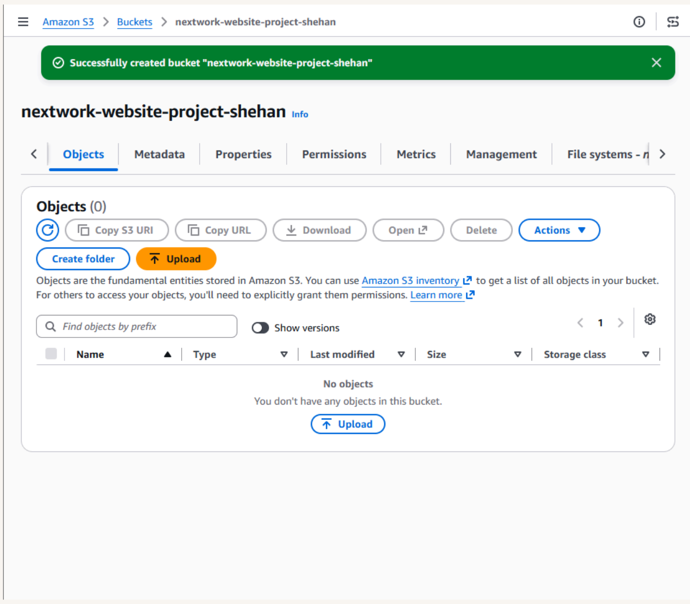
Creating an S3 bucket took me like 10min because i reading and doing it

Region selection

The Region I picked for my S3 bucket was Asia Pacific because i live in Sri Lanka

Understanding bucket name uniqueness

S3 bucket names are globally unique! This means no two buckets across all AWS accounts worldwide can share the same name


Amazon S3 > Buckets > nextwork-website-project-shehan
Successfully created bucket "nextwork-website-project-shehan"
nextwork-website-project-shehan Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - n
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
1
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload

Upload Website Files to S3

What I did in this step

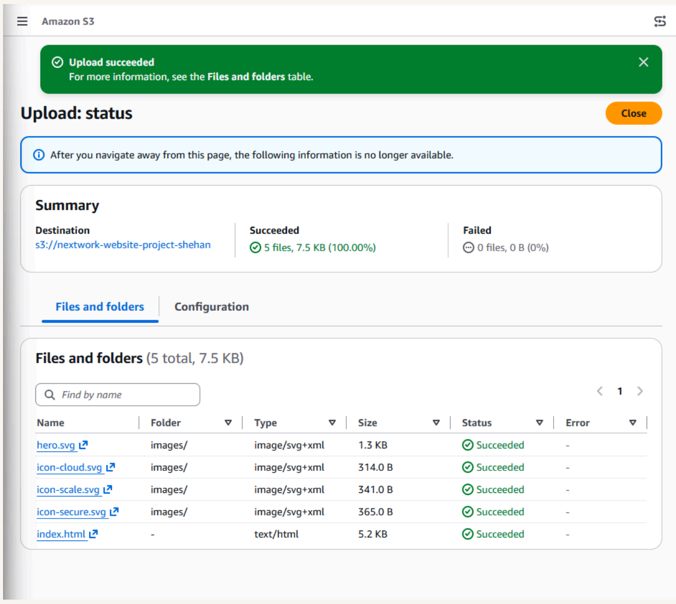
In this step, I will download an HTML file and a zip file of images provided by NextWork.ai and upload both of them into my S3 bucket, because a website needs its content files stored in the cloud before it can be hosted and accessed by anyone through a browser. The HTML file sets up the structure and content of the website while the images make it visual, and by uploading them into the S3 bucket I am making them available to be served as a static website on the internet

Files I uploaded

I uploaded two files to my S3 bucket - they were index.html file and the folder that includes images

How the files work together

Both files are necessary for this project as index.html file is the web page and folder includes the image content that include in the web page


Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://nextwork-website-project-shehan
Succeeded
5 files, 7.5 KB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (5 total, 7.5 KB)
Find by name
1
Name
Folder
Type
Size
Status
Error
hero.svg
images/
image/svg+xml
1.3 KB
Succeeded
-
icon-cloud.svg
images/
image/svg+xml
314.0 B
Succeeded
-
icon-scale.svg
images/
image/svg+xml
341.0 B
Succeeded
-
icon-secure.svg
images/
image/svg+xml
365.0 B
Succeeded
-
index.html
-
text/html
5.2 KB
Succeeded
-

Static Website Hosting on S3

What I did in this step

In this step, I will configure my S3 bucket for static website hosting and visit my public website link, because simply uploading files to an S3 bucket does not automatically make them accessible on the internet. By enabling static website hosting, AWS generates a bucket website endpoint URL that anyone can use to visit my website through a browser, turning my S3 bucket from just a storage space into a fully functioning website host

Understanding website hosting

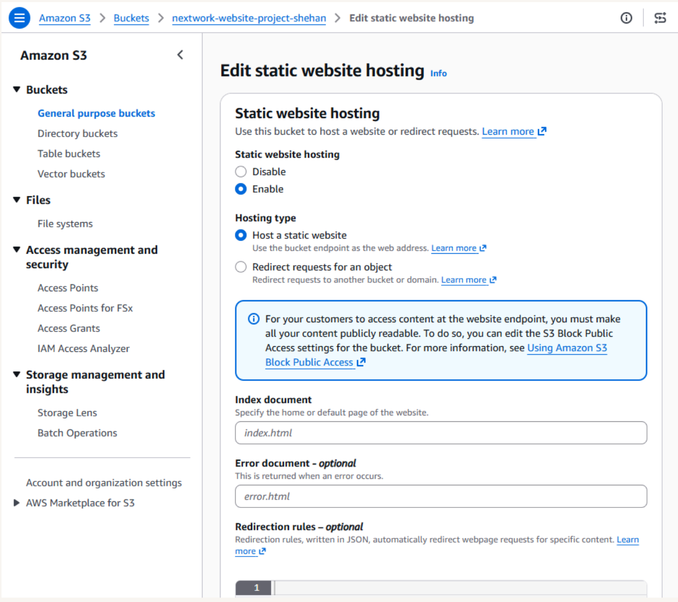
Website hosting is what makes your website public on the internet.

How I enabled website hosting

To enable website hosting with my S3 bucket, I selected the my bucket and went to its properties tab and enable the static web hosting and enter the index document.

Access Control Lists (ACLs)

An ACL is set of rules about who can can access to the resources


Amazon S3 > Buckets > nextwork-website-project-shehan > Edit static website hosting
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
Edit static website hosting Info
Static website hosting
Use this bucket to host a website or redirect requests. Learn more
Static website hosting
Disable
Enable
Hosting type
Host a static website
Use the bucket endpoint as the web address. Learn more
Redirect requests for an object
Redirect requests to another bucket or domain. Learn more
For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access
Index document
Specify the home or default page of the website.
index.html
Error document - optional
This is returned when an error occurs.
error.html
Redirection rules – optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. Learn more
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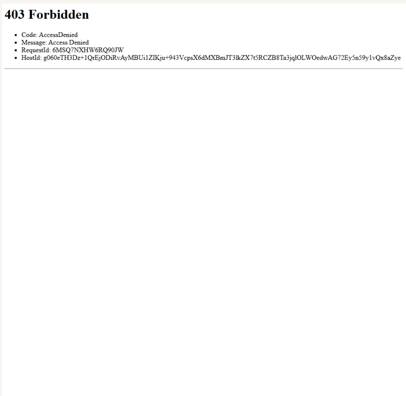
Bucket Endpoints

Understanding bucket endpoint URLs

Once static website is enabled, S3 produces a bucket endpoint URL, which is give access to the static web page

What I saw when I tested the endpoint

When I first visited the bucket endpoint URL, I saw a error message The reason for this error was index file and folder is private by default



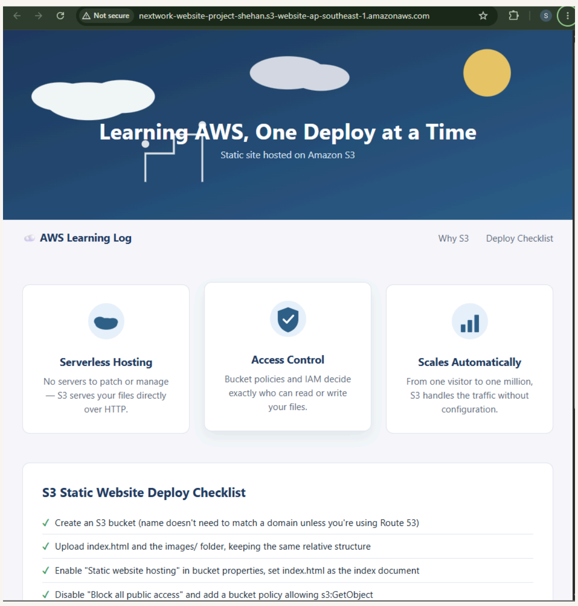
Success!

What I did in this step

In this step, I will make my website files in S3 publicly accessible and see my website live on the internet, because even after enabling static website hosting the files inside the bucket are private by default and cannot be viewed by anyone. By updating the bucket permissions and Access Control List to allow public read access, anyone with the bucket website endpoint URL will be able to open and view my website in their browser, completing the process of hosting a static website on Amazon S3

How I resolved the 403 error

To resolve this 403 Forbidden error, I set public to those index.html and image folder using ACL



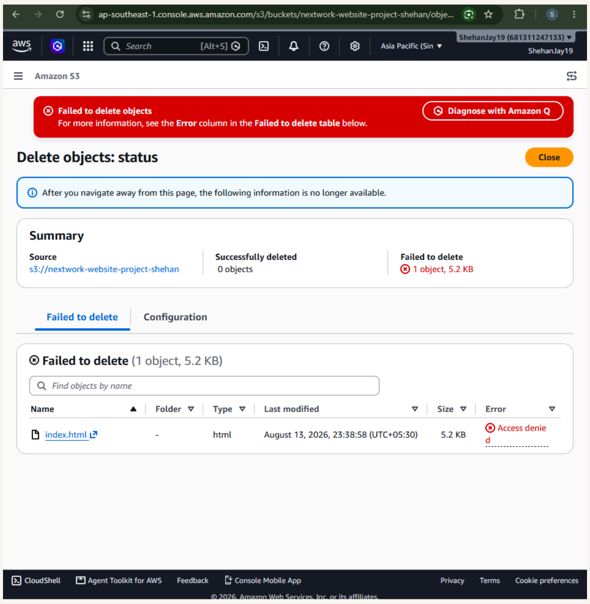
Bucket Policies

What I did in this extension

In this project extension I'm about to set up bucket policy I'm doing this so that people cant delete my index.html file

Understanding bucket policies

An alternative to ACLs are bucket policies, which are JSON-based rules attached directly to an S3 bucket that define who can access the bucket and what actions they are allowed to perform on it. The benefit of using bucket policies is that they give you more detailed and flexible control over access permissions, allowing you to set rules for entire buckets at once, restrict access by IP address, require specific conditions, and manage permissions in one central place, while ACLs are useful for quickly setting basic read and write permissions on individual files or buckets, especially when you need to make a specific file publicly accessible without changing the whole bucket policy



What my bucket policy does

My bucket policy allows public read access to all files inside my S3 bucket by granting the s3:GetObject permission to everyone, which means any person on the internet can view and download the files stored in my bucket through the browser. I tested this by copying my bucket website endpoint URL and opening it in a new browser tab, and saw my website loading successfully on the internet with all the HTML content and images displaying correctly, confirming that the bucket policy was working as expected

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